Univariate Analysis

What Is Univariate Analysis?

"Univariate" simply means **"one variable."** Univariate analysis is the practice of looking at **a single column of data at a time** to understand its basic behavior - before you start comparing it to anything else.

This guide covers three of the most common univariate charts:

1. **Histogram** - for numeric data (numbers you can measure, like order value or delivery time)
2. **Bar Chart** - for categorical data (labels or groups, like city or gender)
3. **Boxplot** - for spotting spread and unusual values in numeric data

For each chart below, we'll first walk through **how the chart works in general**, using simple examples, so you can focus purely on reading the chart itself. Then we'll show **what it looks like when we apply it to UrbanCart's data**, with the real business insights it uncovers.

Before diving in, here are a few terms we'll use throughout this guide, explained in plain English:

- **Distribution**: This just means "how the values are spread out." If you lined up a group of people from shortest to tallest, the *shape* of that lineup is the distribution.
- **Skewness**: A distribution is "skewed" when the data isn't evenly balanced around the middle-it leans heavily toward one side, with a few extreme values pulling the average far off to the other.
    - **Right-skewed (Positive skew):** Most of the data piles up on the lower/smaller end, but a few unusually high values stretch out to the right.
        - *Example:* Income. Most people earn a typical salary, but a small handful of billionaires pull the average way up.
    - **Left-skewed (Negative skew):** Most of the data piles up on the higher/larger end, but a few unusually low values stretch out to the left.
        - *Example:* Age at retirement. Most people retire in their 60s or 70s, but a few people retire much earlier, in their 20s or 30s.
- **Outliers**: These are values that sit far away from the rest of the data - unusually high or unusually low. For example, if most people finish a 5k run in 25-35 minutes but one finishes in 120 minutes, that 120-minute result is an outlier.
- **Quartiles**: If you sort your data from smallest to largest and cut it into four equal-sized groups, the cut points are called quartiles (quarter => quartiles).

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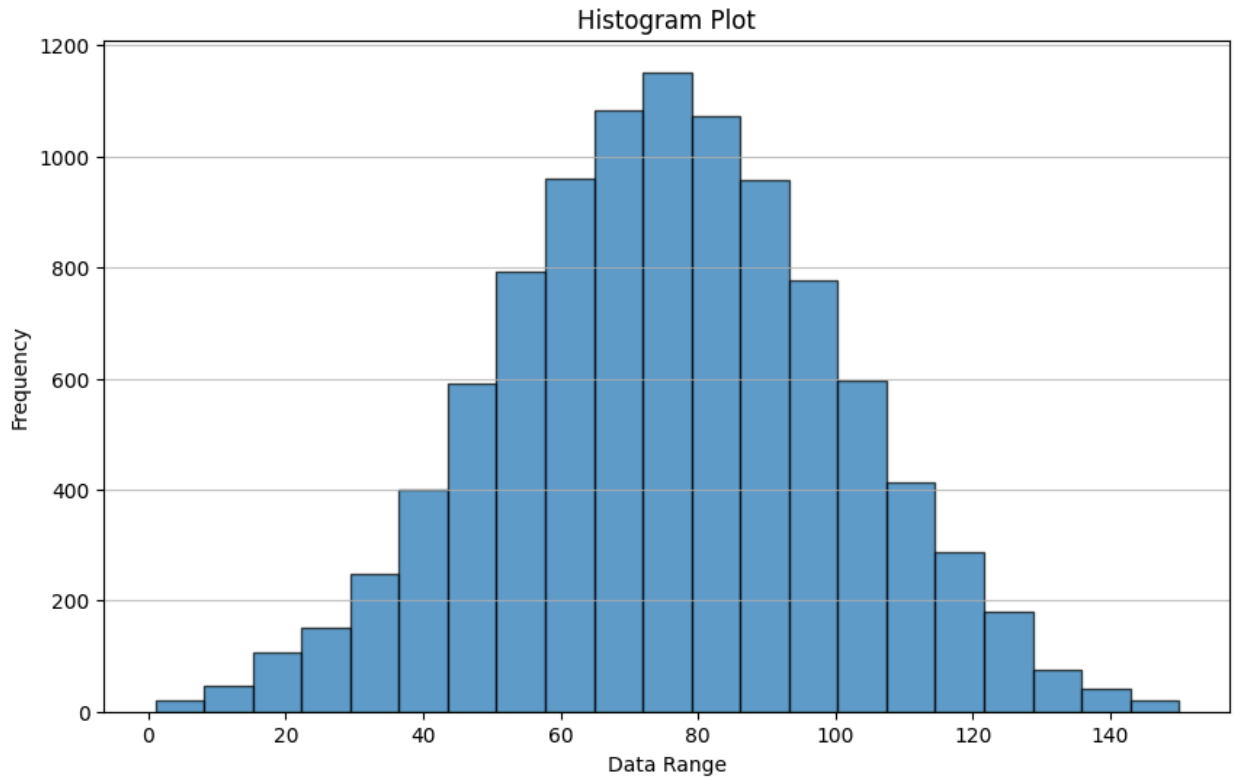
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1. Histogram

Part 1: The Basics

What Is a Histogram?

A **histogram** takes a numeric column (like a person's age, or the price of a product) and splits it into small ranges called "bins" (for example, ages 0-10, 10-20, 20-30). It then counts how many records fall into each bin and draws a bar for each one. The taller the bar, the more records fall in that range.



Understanding the Axes & Visuals

- **X-axis**: The range of values for whatever numeric measurement you're studying (for example, a person's age from 0 to 100).
- **Y-axis**: How many records fall into each range - this is the "count" or "frequency."
- **Bars**: Each bar represents one "bin," or range of values. Taller bars mean more records fall in that range.

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When and Why to Use It

Use a histogram whenever you want to answer questions like:

- "Where is the main peak of the data located?"
- "Are most values clustered close together, or spread out across a wide range?"
- "Is there one common range where most values sit, or several separate clusters?"

It's the go-to chart whenever you have a numeric column and want to understand its overall shape - is it balanced, lopsided, or spread across a wide range?

General Insights

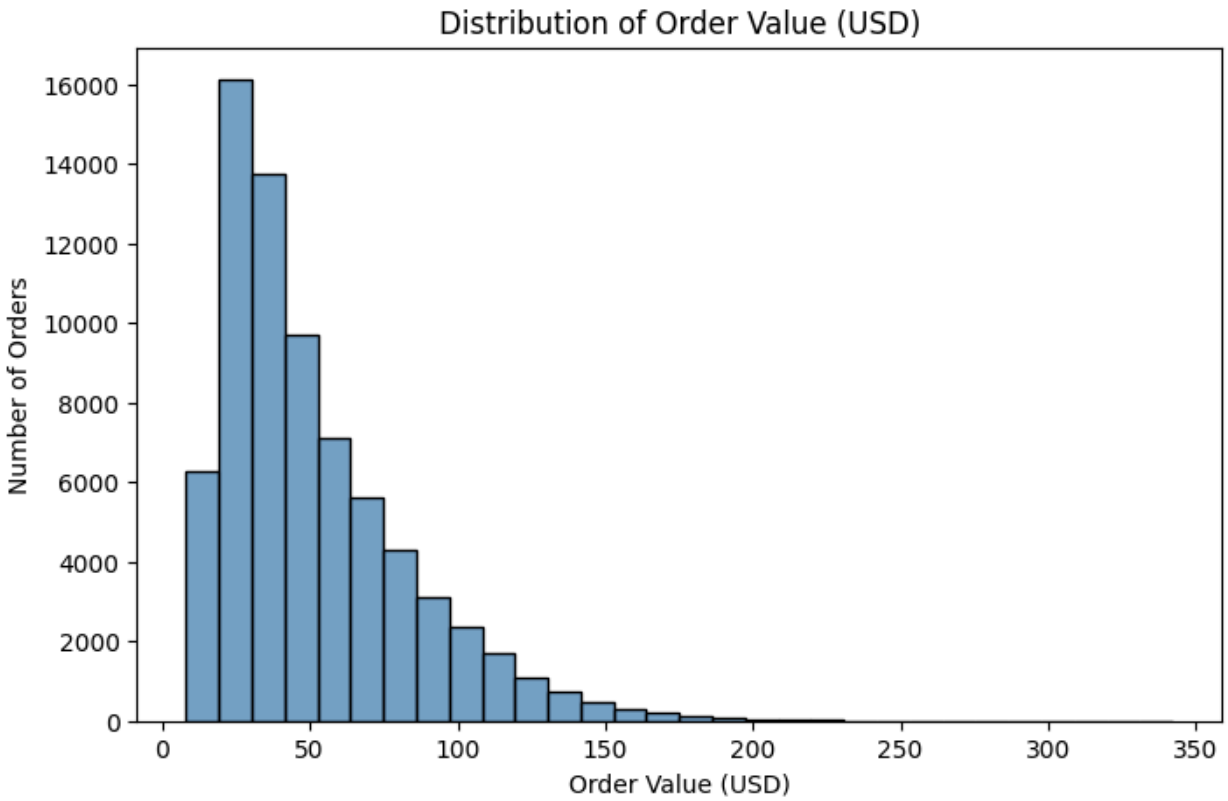
- **Shape**: Is the data roughly symmetric (a balanced hill) or skewed (leaning one way with a long tail)?
- **Peaks**: Is there one common range where most values cluster (one hill), or are there two or more separate clusters (multiple hills), suggesting different underlying groups?
- **Spread**: Is the data tightly packed in a narrow range, or widely spread out?
- **Outliers**: Are there a few bars far off to the side, isolated from the rest of the data?

Part 2: Applying It to UrbanCart

Now let's use this same chart on UrbanCart's own data. Instead of a generic numeric column, we can plot something like `order_value_usd` (from `orders.csv`) to see how customer spending is actually shaped.

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Case-Specific Insights

- For `order_value_usd`: We see that the distribution is skewed toward the low end with a long tail of high-value orders; it suggests most customers place modest orders, while a smaller group of "big spenders" drives up the average - useful for identifying a high-value customer segment.

For other attributes in the data, one can use a boxplot for the following:

- For `promised_time_min` or `actual_time_min` (from `deliveries.csv`): A histogram can reveal whether UrbanCart's delivery times cluster tightly around a "typical" promise, or whether there's a wide spread that might explain inconsistent customer experiences.
- For `discount_amount_usd`: A peak near zero with a long tail could indicate that most orders receive small or no discounts, while a few receive large discounts - worth checking against `promo_used` and retention behavior later in bivariate analysis.

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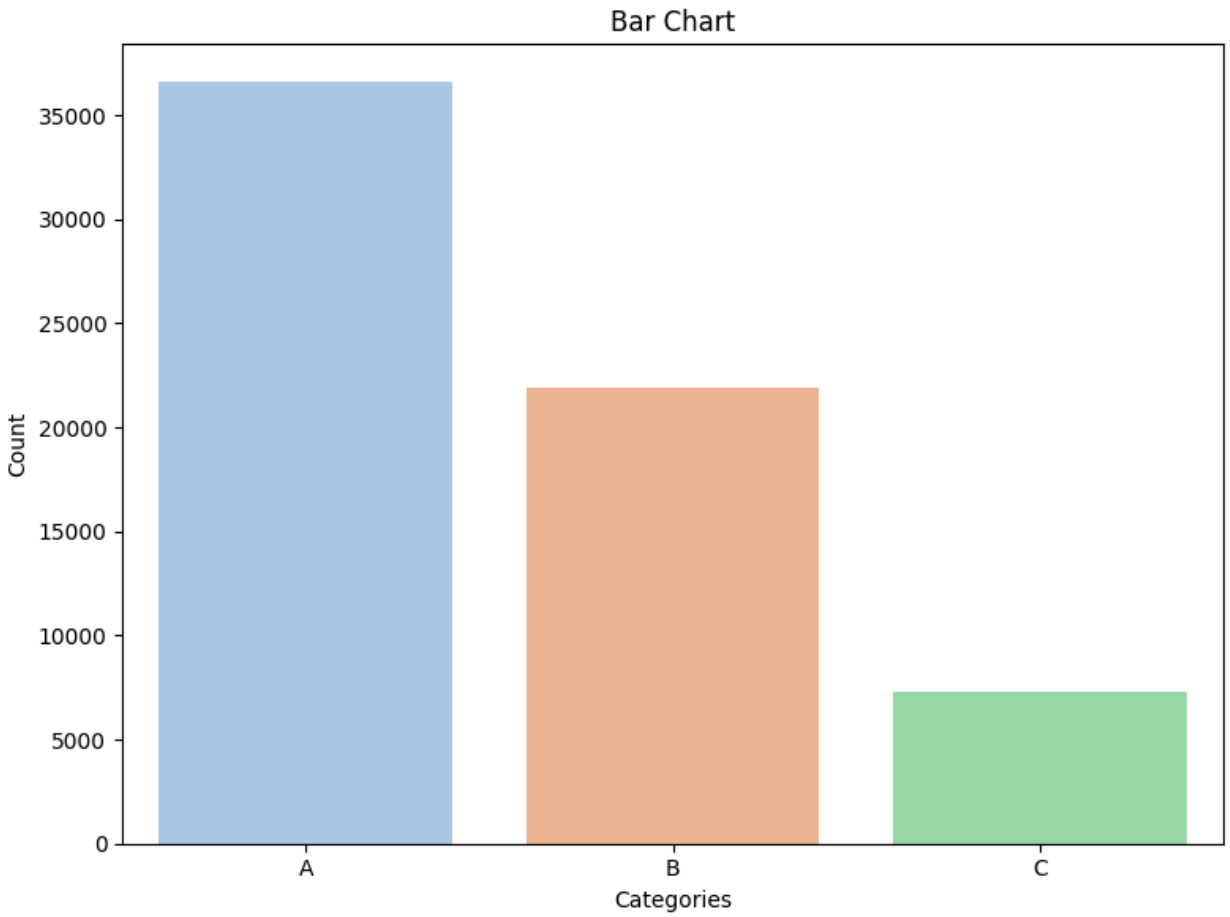
2. Bar Chart

Part 1: The Basics

What Is a Bar Chart?

A **bar chart** is used for **categorical data** - columns made up of labels or groups rather than numbers, such as favorite fruit, car brand, or shirt size. Instead of showing ranges of numbers like a histogram, it simply counts how many records fall into each category and draws one bar per category.

Think of it as a tally: "How many people picked apples? How many picked oranges? How many picked bananas?" Each answer gets its own bar.



Understanding the Axes & Visuals

- **X-axis**: The categories themselves (e.g., A, B, C).
- **Y-axis**: The count - how many records belong to each category.
- **Bar height**: Taller bars mean that category appears more often in the dataset.

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- **Bar order**: Bars can be sorted alphabetically, by size (largest to smallest), or in a custom order - sorting by size often makes patterns easier to spot.

When and Why to Use It

Use a bar chart whenever you want to answer questions like:

- "How many records fall into each category?"
- "Which category is the most common, and which is the rarest?"
- "Is the data evenly split across categories, or dominated by one group?"

It's the standard tool for understanding how a labeled/grouped column breaks down - essentially a visual version of counting how many times each label appears.

General Insights

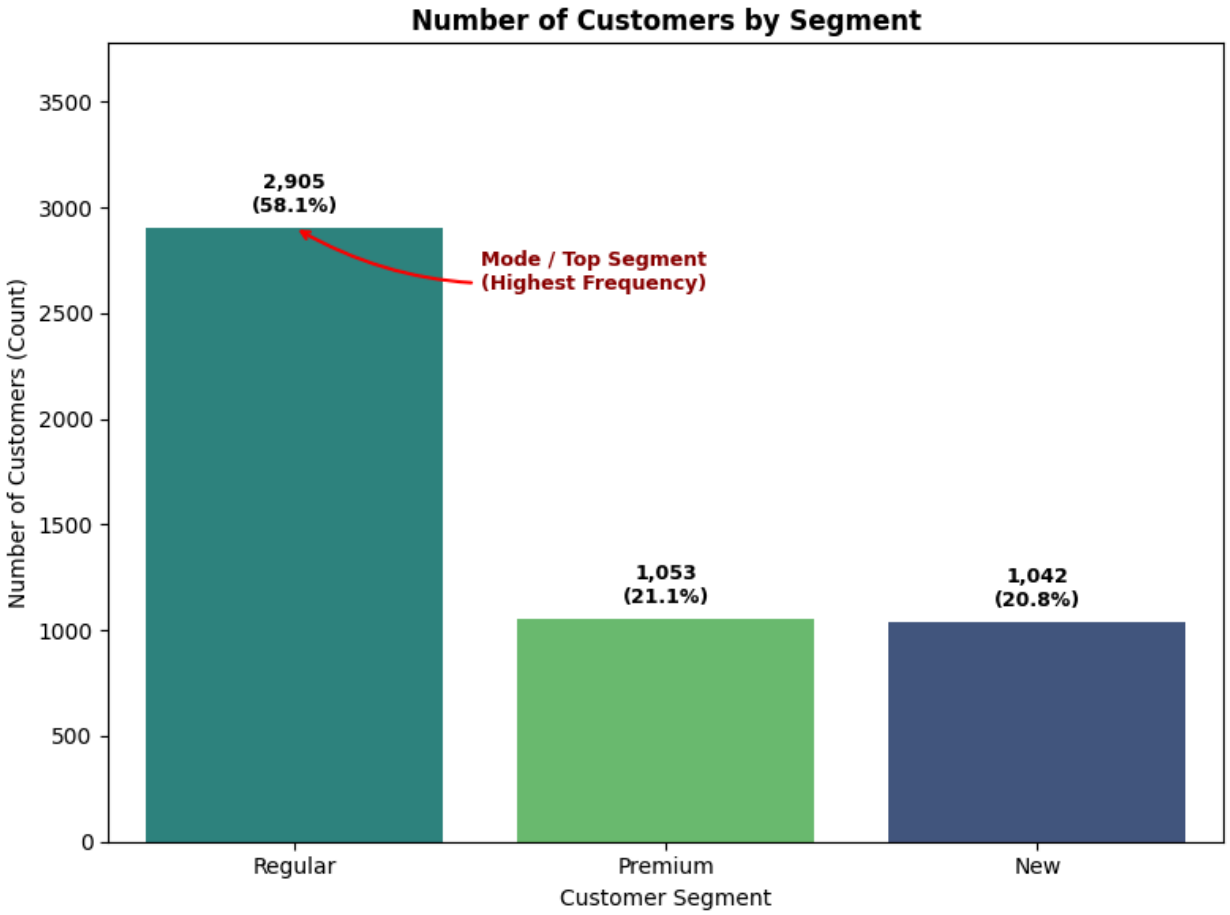
- **Dominant categories**: Is one category far more common than the others, suggesting an imbalance?
- **Balance**: Are the categories roughly evenly split, suggesting a diverse or well-distributed group?
- **Rare categories**: Are there categories with very few records that might need to be grouped together or treated with caution in later analysis (since small groups can produce unreliable patterns)?

Part 2: Applying It to UrbanCart

Now let's apply this to UrbanCart's data. Instead of counting fruit, we can count how many customers fall into each `customer_segment` (from `customers.csv`) - New, Regular, or Premium.

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Case-Specific Insights

- For `customer_segment`: "Regular" customers dominate while "Premium" customers are a small slice, it tells UrbanCart that premium upgrades may be an untapped growth opportunity.

For other attributes in the data, one can use a bar chart for the following:

- For `preferred_device`: If the vast majority of customers use the Mobile App over Web, this could guide where UrbanCart invests in retention features (e.g., app-based loyalty perks).
- For `delivery_type` (from `orders.csv`): If "On-Demand" heavily outweighs "Scheduled" and "Express," it suggests customer preference for immediacy, which could shape staffing and vendor partnership decisions.
- For `vendor_category`: Seeing whether "Restaurant" orders vastly outnumber "Pharmacy" or "Retail" orders helps UrbanCart understand which vendor types drive the bulk of platform activity.

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3. Boxplot

Part 1: The Basics

What Is a Boxplot?

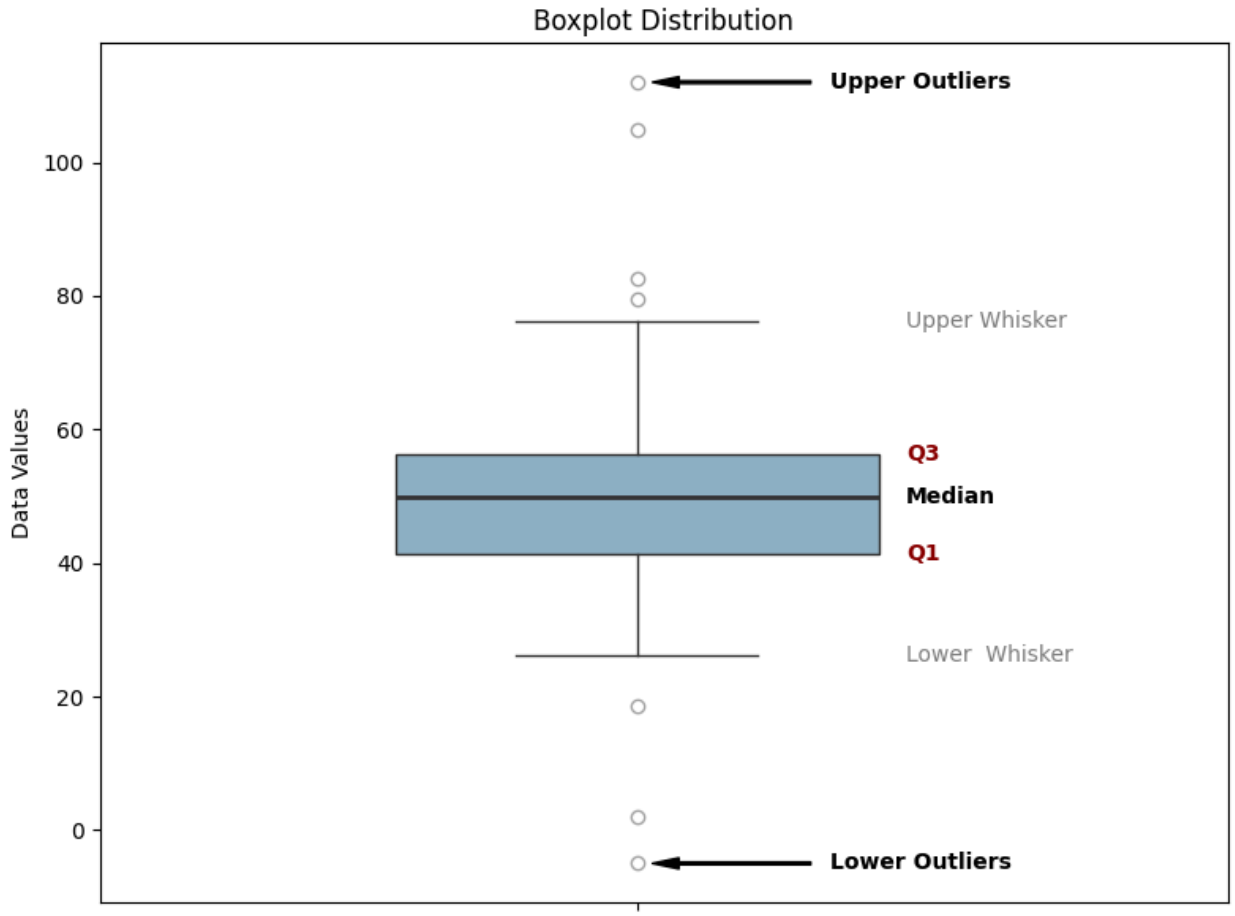
A **boxplot** (sometimes called a "box-and-whisker plot") is a compact way to summarize a numeric column using five key landmarks: the lower whisker, the first quartile, the median, the third quartile, and the upper whisker (excluding extreme outliers, which are shown separately as dots).

In plain English:

- The **box** represents the "middle 50%" of your data - the most typical range of values.
- The **line inside the box** is the **median** - the value exactly in the middle when all data points are sorted from smallest to largest (not to be confused with the average, which can be pulled around by extreme values).
- The **whiskers** (the lines extending from the box) show the reasonable range of values outside the box.
- **Dots beyond the whiskers** are **outliers** - unusually high or low values that stand apart from the rest of the data.
- **IQR (Interquartile Range)**: This is the "width" of the box, representing the range covered by the middle 50% of your data. It's calculated by subtracting the first quartile from the third quartile (IQR = Q3 - Q1).
- **Whiskers**: The upper and lower whiskers are calculated as 1.5 times the IQR added to or subtracted from the quartiles (*Q3 + 1.5 * IQR* for the Upper Whisker*; Q1 - 1.5 * IQR* for the Lower Whisker). It should be noted that the whiskers do not necessarily extend to these calculated boundaries. In standard boxplot implementations, the whisker extends to the furthest data point that is within the calculated boundary (1.5 * IQR). If no data point reaches the boundary, the whisker **stops** at the **actual maximum or minimum** observed data point.
- Data points that fall beyond these whisker lines are considered outliers.

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Understanding the Axes & Visuals

- **X-axis**: Either left blank (for a single boxplot) or used to show categories side-by-side (e.g., one box per team).
- **Y-axis**: The numeric value being examined (e.g., commute time, test score).
- **Box**: Spans from the first quartile (25% mark) to the third quartile (75% mark) - the middle half of the data.
- **Line in the box**: The median - the true "middle" value.
- **Whiskers**: Extend to the smallest and largest values that are not considered outliers.
- **Individual dots**: Outliers - data points far from the rest of the pack.

When and Why to Use It

Use a boxplot whenever you want to answer questions like:

- "What is the typical value, and are there any unusually extreme values?"
- "How spread out is the data, and are there a few unusually high or low points?"
- "Are there unusual, extreme values that might indicate data errors or rare events?"

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Boxplots are especially useful for quickly spotting outliers and comparing the spread (not just the average) of numeric data.

General Insights

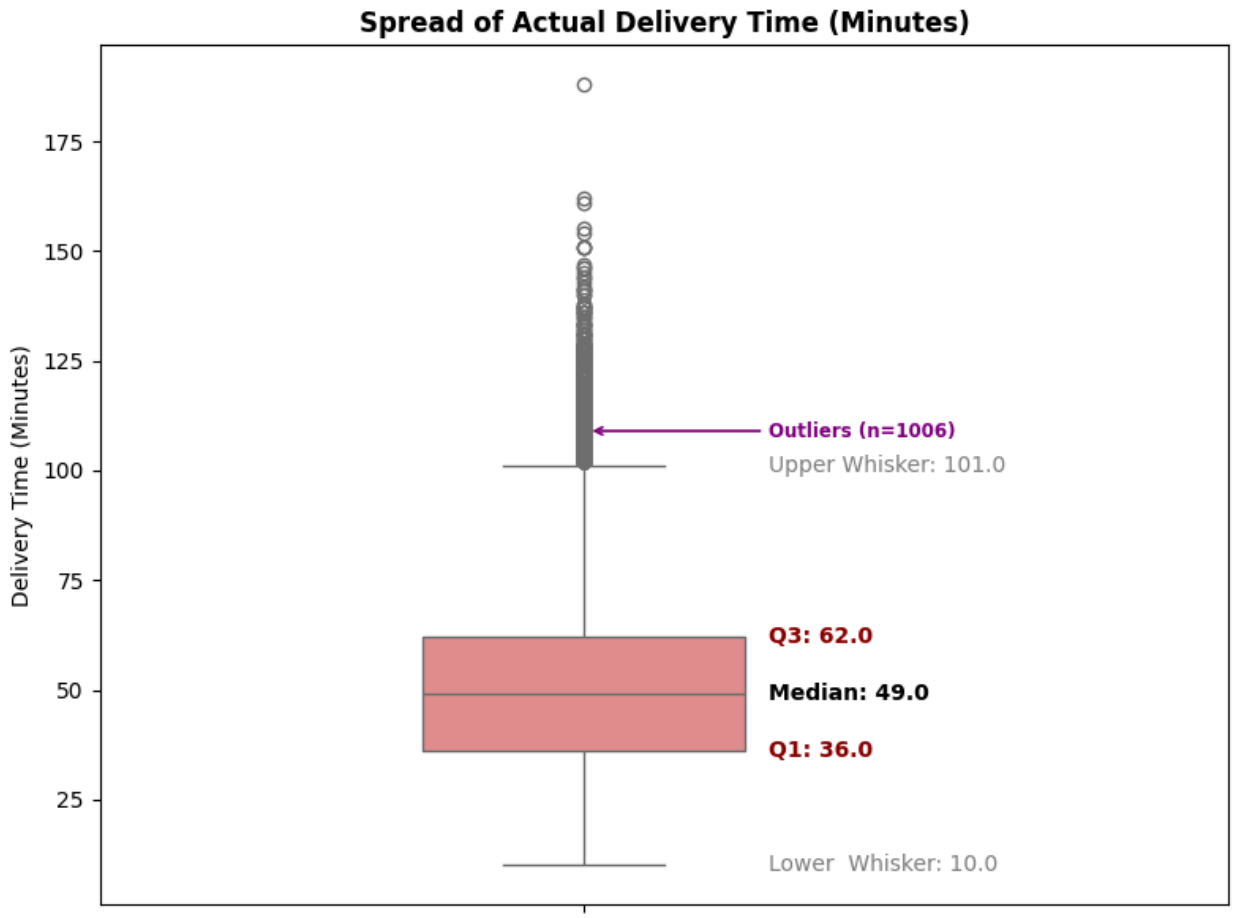
- **Spread**: A tall box means the values are widely spread; a short box means values are tightly clustered.
- **Median position**: If the median line sits closer to one edge of the box than the other, the data is skewed in that direction.
- **Outliers**: Individual dots beyond the whiskers flag unusual cases worth investigating - are they data entry errors, or genuine rare events?
- **Symmetry**: Roughly equal whisker lengths on both sides suggest a balanced spread; very unequal whiskers suggest skewness.

Part 2: Applying It to UrbanCart

Now let's use a boxplot on UrbanCart's data. For example, we can look at `actual_time_min` (from `deliveries.csv`) to see how delivery times are spread out, and whether any deliveries took an unusually long time.

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Case-Specific Insights

- For `actual_time_min`: The boxplot reveals that most deliveries take a standard amount of time (a small box near 50 minutes), while a handful of dots show extreme delivery times - these extreme cases may be worth investigating operationally, since they can heavily influence customer satisfaction.

For other attributes in the data, one can use a boxplot for the following:

- For `order_value_usd`: A boxplot can show whether the "typical" order sits in a narrow price band, with a few dots representing unusually large bulk orders.
- For `items_count`: Comparing the box and whiskers can show whether most orders contain a small, predictable number of items, or whether basket sizes vary widely across the customer base.
- For `customer_rating`: Since ratings only range from 1 to 5, a boxplot can quickly show whether customers tend to rate deliveries highly (a box clustered near 4-5) or whether ratings are more spread out, signaling inconsistent service experiences.

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Quick Recap

Chart	Best For	Key Question It Answers
Histogram	Numeric columns	"What does the overall shape of this number look like?"
Bar Chart	Categorical columns	"How many records fall into each group or label?"
Boxplot	Numeric columns	"Where is the typical range, and are there unusual extreme values?"

Univariate analysis is always the **first step** in exploring data - it builds the foundation before moving on to comparing two variables (bivariate analysis) or several variables at once (multivariate analysis).

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